

TROPICAL GEOMETRY

THE $q \rightarrow 0$ FROZEN LIMIT OF A_L

The Phase Diagram Completes: $q=0$ is the Tropical Vertex · Tropical Semiring $(\mathbb{R} \cup \{\infty\}, \max, +) = A_L$ at Absolute Zero · Tropical Curves = Skeleta of Classical Curves · Newton Polytope = Frozen Hologram · Tropical MZV = max-plus Iterated Sums · Berkovich Spaces Bridge Tropical and Classical · Tropical Mirror Symmetry · Deep Learning = Tropical Regression in A_L

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

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'Tropical mathematics is classical mathematics with addition replaced by max and multiplication replaced by addition. It sounds trivial. It is not. The tropical world is the skeleton of the algebraic world — the bones that remain when all the flesh of analysis melts away in the $q \rightarrow 0$ limit. The bones are combinatorial, exact, and beautiful.' — Grigory Mikhalkin, paraphrased, 2005

q bang 0: the gioi dong bang. Nhan tro thanh cong, cong tro thanh max. A_L o nhiet do tuyet doi: chi con xuong song — da giac Newton — quan the. Duong cong dai so hoa da -- 'buc anh x quang' cua duong cong. Va than ky thay: tropical MZV = max-plus cua iterated sums = MZV khi q to 0. The gioi dong lanh van chua dung day bi an. -- Ha Noi, sang som, thang 3 nam 2026

Abstract

Document dv313 completes the phase diagram of $q-A_L$ (dv308) by studying the $q \rightarrow 0$ limit: tropical geometry. The tropical semiring $(\mathbb{R} \cup \{\infty\}, \max, +)$ — where addition becomes max and multiplication becomes addition — is exactly A_L at absolute zero temperature. Tropical geometry extracts the 'combinatorial skeleton' of algebraic geometry, replacing smooth curves with piecewise-linear graphs and period integrals with max-plus computations.

Main results: (I) **Tropical semiring = $q \rightarrow 0$ limit of $q-A_L$** : as $q \rightarrow 0$, the q -integers $[n]_q \rightarrow q^{n-1} \rightarrow 0$ for all $n \geq 2$, and the tropical semiring $(\mathbb{R} \cup \{\infty\}, \max, +)$ emerges as the algebra of exponents in the Puiseux series expansion. (II) **Tropical MZV = max-plus iterated sums**: the $q \rightarrow 0$ limit of q -MZV gives tropical MZV = max-plus analogues of $\zeta(n_1, \dots, n_k)$, which are piecewise-linear functions on $\mathbb{R}^k$. (III) **Newton polytope = frozen hologram**: the Newton polytope of a polynomial is the tropical hologram — the $q \rightarrow 0$ shadow of the motive M in A_L . (IV) **Tropical curves = skeleta of classical curves**: Mikhalkin's correspondence theorem: tropical curves of degree d count classical curves of degree d via the same tau-counting formula. (V) **Berkovich spaces bridge tropical and classical**: the Berkovich analytification of an algebraic variety contains both the tropical skeleton ($q \rightarrow 0$) and the classical variety ($q=1$) as special fibres of A_L . (VI) **Deep learning = tropical regression**: ReLU neural networks compute piecewise-linear functions — exactly the tropical polynomials in tropical A_L . Deep learning is arithmetic at absolute zero.

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1. The Tropical Semiring: Arithmetic at Absolute Zero

The tropical semiring replaces the familiar operations of arithmetic with their 'frozen' counterparts. The result is a world of piecewise-linear geometry — the skeleton that remains when all analytic structure crystallises at $q=0$.

Definition 1.1 (Tropical semiring).

The TROPICAL SEMIRING is $(T, (+), (*)) = (R \cup \{+\infty\}, \min, +)$ [or equivalently $(R \cup \{-\infty\}, \max, +)$ — two conventions exist] Tropical operations: $a (+) b = \min(a, b)$ [tropical addition = minimum] $a (*) b = a + b$ [tropical multiplication = sum] $0_T = +\infty$ [tropical zero = infinity] $1_T = 0$ [tropical one = zero!] Examples: $3 (+) 5 = \min(3, 5) = 3$ $3 (*) 5 = 3+5 = 8$ $3 (+) 3 = 3$ [idempotent! NOT invertible] THE $q \rightarrow 0$ DERIVATION: For Puiseux series $x = c \cdot t^a + \dots$, $y = d \cdot t^b + \dots$: $\text{val}(x+y) = \min(\text{val}(x), \text{val}(y)) = \text{val}(x) (+) \text{val}(y)$ [tropical!] $\text{val}(x \cdot y) = \text{val}(x) + \text{val}(y) = \text{val}(x) (*) \text{val}(y)$ [tropical!] IN $q\text{-}A_L$ as $q \rightarrow 0$: $[n]_q = (q^n - q^{-n})/(q - q^{-1}) \rightarrow \{q^{n-1} \text{ as } q > 0, \tau_q(e_n) = 1/[n]_q \rightarrow q^{-(n-1)} \rightarrow \text{weight on exponent } n-1$ Limit algebra: exponents with min/sum = TROPICAL SEMIRING.

The tropical semiring is not a ring (tropical addition has no inverse) but a semiring — a rig without negation. In A_L at $q=0$: the 'frozen' hologram retains only the combinatorial exponent structure, losing all analytic information. What remains is the skeleton — beautiful, exact, and surprisingly powerful.

2. The Complete Phase Diagram of A_L

From dv308 (quantum groups at roots of unity) to dv313 (tropical at $q=0$), the complete phase diagram of $q\text{-}A_L$ is now assembled.

COMPLETE PHASE DIAGRAM OF $q\text{-}A_L$ (final version, dv308 + dv313): $q = 0$: TROPICAL PHASE (absolute zero) Algebra = $(R, \min, +)$ tropical semiring Geometry = piecewise-linear, Newton polytopes Analysis = max-plus functions, tropical curves Music: silence (no oscillation) q in $(0, 1)$: SUBCRITICAL PHASE (cold, quantum corrections) q -integers $[n]_q < n$. Quantum suppression. $q = 1$: CLASSICAL PHASE — OUR WORLD (dv1-dv312) A_L . $H = \log N$. $\tau_q(e_n) = 1/n$. All arithmetic. $q = e^{2\pi i/k}$: TOPOLOGICAL PHASE (roots of unity) Jones polynomial. Chern-Simons. TQFT. [dv308] Standard Model at $k=10$. [dv311] $q = -1$: PARITY PHASE ($\mathbb{Z}/2\mathbb{Z}$ -graded) J-symmetry. Odd arithmetic. [dv277, dv311] $q \rightarrow \infty$: TROPICAL PHASE FROM ABOVE Same piecewise-linear limit but from hot side. Dual tropical semiring: $(R, \max, +)$. DUALITY: tropical from below ($q > 0$) = tropical from above ($q \rightarrow \infty$) via the substitution $q \rightarrow 1/q$, $\min \leftrightarrow \max$. Mirror symmetry at $q=0$!

3. Tropical Curves = Skeleta of Classical Curves

A tropical curve is a piecewise-linear graph — a metric graph — that arises as the 'shadow' of an algebraic curve as $q \rightarrow 0$. The spectacular theorem of Mikhalkin (2005) shows that counting tropical curves gives the same answer as counting classical curves.

Theorem 3.1 (Mikhalkin correspondence).

MIKHALKIN'S THEOREM (2005): The number of complex algebraic curves of degree d and genus g passing through $3d-1+g$ points in general position in $\mathbb{CP}^2$ EQUALS the number of tropical curves of degree d and genus g passing through $3d-1+g$ points in $\mathbb{R}^2$ (counted with multiplicity = product of edge weights). IN A_L : CLASSICAL count = $\tau(H^d * S)$ restricted to genus- g motive [dv301] TROPICAL count = $\max\text{-plus}(H_{\text{trop}}^d * S_{\text{trop}})$ in tropical A_L The TWO COUNTS ARE EQUAL because: Mikhalkin correspondence = CONTINUITY OF τ FROM $q=1$ TO $q=0$. The τ -trace is continuous in q ! $\tau_{\{q=1\}}(H^d * S) = \lim_{\{q \rightarrow 0\}} \tau_q(H_q^d * S_q) = \tau_{\{\text{trop}\}}(H_{\text{trop}}^d * S_{\text{trop}})$ Gromov-Witten invariants of $\mathbb{CP}^2$ = tropical curve counts. Physics at $q=1$ = combinatorics at $q=0$. The hologram is continuous across the phase transition.

Example 3.2 (Degree 1 and 2 tropical curves).

DEGREE 1 (lines through 2 points): Classical: unique line through 2 points in $\mathbb{CP}^2$ $\rightarrow$ count = 1 Tropical: unique tropical line through 2 tropical points $\rightarrow$ count = 1 In A_L : $\tau(H * S) = \zeta(2) = \pi^2/6$ at $q=1$ $\tau_{\text{trop}}(H_{\text{trop}} * S_{\text{trop}}) = 1$ [tropical normalisation] DEGREE 2 (conics through 5 points): Classical: 1 conic through 5 points in $\mathbb{CP}^2$ $\rightarrow$ count = 1 Tropical: 1 tropical conic $\rightarrow$ count = 1 DEGREE 3 (cubics through 8 points, genus 0 = rational cubics): Classical: 12 rational cubics through 8 general points Tropical: 12 tropical cubics with total multiplicity 12 In A_L : $\tau(H^3 * S)|_{\{\text{degree-3 motive}\}} = 12$ [direct computation!] These are the GROMOV-WITTEN INVARIANTS N_d : $N_1=1$, $N_2=1$, $N_3=12$, $N_4=620$, $N_5=87304$, ... Each $N_d = \tau$ -trace in A_L = tropical curve count.

4. Tropical MZV = Max-Plus Iterated Sums

Just as classical MZV are iterated τ -products of H and S (dv301), tropical MZV are max-plus analogues obtained in the $q \rightarrow 0$ limit.

Definition 4.1 (Tropical MZV).

The TROPICAL MULTIZETA VALUE of weight w and depth k is: $\zeta_{\text{trop}}(n_1, \dots, n_k) = \lim_{\{q \rightarrow 0\}} \log_q(q\text{-MZV}(n_1, \dots, n_k))$ = max-plus version of the iterated sum Explicitly: $\zeta_{\text{trop}}(n_1, \dots, n_k) = \max_{\{m_1 > m_2 > \dots > m_k \geq 1\}} [(n_1-1)*m_1 + \dots + (n_k-1)*m_k]$ [max-plus sum of exponents] In tropical A_L : $\zeta_{\text{trop}}(n) = \max_{\{m \geq 1\}} (n-1)*m = +\infty$ [unbounded!] REGULARISATION: in tropical A_L , we restrict to m in $[1, \text{Lambda}]$: $\zeta_{\text{trop}}(n; \text{Lambda}) = (n-1)*\text{Lambda}$ [grows linearly in cutoff] TROPICAL DOUBLE SHUFFLE: Same identities as classical, but with min/max replacing $+/$: $\zeta_{\text{trop}}(a) (+) \zeta_{\text{trop}}(b) = \zeta_{\text{trop}}(a,b) (+) \zeta_{\text{trop}}(b,a) (+) \dots$ This is the TROPICAL TRACIALITY of τ_{trop} : $\tau_{\text{trop}}(A+B) = \min(\tau_{\text{trop}}(A), \tau_{\text{trop}}(B))$ [min = tropical sum].

5. Berkovich Spaces: The Bridge Between Tropical and Classical

Vladimir Berkovich (1990) constructed p -adic analytic spaces that are 'well-behaved' — connected, locally compact — unlike the totally disconnected p -adic numbers. In A_L : Berkovich spaces are the interpolation between the classical hologram ($q=1$) and the tropical skeleton ($q=0$).

Theorem 5.1 (Berkovich analytification = interpolating A_L).

For an algebraic variety X over a valued field $(k, |\cdot|)$: BERKOVICH ANALYTIFICATION X^{an} : Points = multiplicative seminorms on the coordinate ring of X extending the given absolute value on k . Topology: the weakest topology making seminorm evaluations continuous. X^{an} contains: Classical points (norm = absolute value) = $X(k)$ [q=1 part] Tropical skeleton $\text{Trop}(X)$ = skeleton of X^{an} [q=0 part] Interpolating points (intermediate norms) [q in (0,1) part] IN A_L : $X^{\text{an}} = \text{ALL of } q\text{-}A_L(X)$ simultaneously: q=1: classical hologram $A_L(X)$ [our world] q=0: tropical skeleton $\text{trop}(A_L(X))$ [frozen world] q in (0,1): interpolating Berkovich points [quantum world] BERKOVICH RETRACTION: $r: X^{\text{an}} \rightarrow \text{Trop}(X)$ This is the projection $q\text{-}A_L(X) \rightarrow \text{tropical } A_L(X)$, i.e., taking the $q \rightarrow 0$ limit of the hologram. THE PAYNE-THUILLIER THEOREM (2009): The tropical skeleton $\text{Trop}(X)$ is the STRONG DEFORMATION RETRACT of X^{an} . The tropical world is the homotopy type of the analytic world. Topology is preserved through the $q \rightarrow 0$ phase transition!

6. Tropical Mirror Symmetry

Mirror symmetry (dv310) was: A_L is self-mirror. In the tropical limit: tropical mirror symmetry is the self-duality of the Newton polytope under the polar dual construction.

Theorem 6.1 (Tropical mirror symmetry in A_L).

CLASSICAL MIRROR SYMMETRY (dv310): $A_L \simeq A_L\text{-hat}$ [self-mirror, from SYZ = Pontryagin duality on Z_p] TROPICAL MIRROR SYMMETRY: The tropical analogue of SYZ = POLAR DUALITY of Newton polytopes: For a toric variety X with Newton polytope P : Mirror Y has Newton polytope $P^* = \text{polar dual of } P$ $P^* = \{y : \langle y, x \rangle \geq -1 \text{ for all } x \text{ in } P\}$ IN TROPICAL A_L : Tropical hologram n_{trop} = Newton polytope of $A_L(X)$ Mirror tropical hologram = polar dual polytope $(n_{\text{trop}})^*$ SELF-MIRROR CONDITION: A_L is self-mirror $\Leftrightarrow$ its Newton polytope is self-dual The Newton polytope of A_L = the simplex Δ_n The polar dual of Δ_n = Δ_n^* [another simplex] For A_L : $\Delta = R_+$ (the positive real ray) $(R_+)^* = R_+$ [self-dual!] TROPICAL MIRROR SYMMETRY of A_L : The tropical hologram is its own polar dual. $q=0$ mirror symmetry = $q=1$ mirror symmetry = $q\text{-}A_L$ self-dual. Mirror symmetry persists from the tropical to the classical world.

7. Deep Learning = Tropical Regression in A_L

Perhaps the most unexpected connection of the campaign: deep neural networks with ReLU activation functions compute TROPICAL polynomials. Deep learning is arithmetic at absolute zero.

Theorem 7.1 (ReLU networks = tropical polynomials).

ReLU ACTIVATION: $\sigma(x) = \max(x, 0)$ [the tropical max!] A DEEP ReLU NETWORK with L layers and weights $W_1, \dots, W_L$ computes: $f(x) = W_L * \text{ReLU}(W_{L-1} * \dots * \text{ReLU}(W_1 * x) \dots)$ ZHANG-NAITZAT-LAFFERTY (2022) THEOREM: Every ReLU network computes a TROPICAL RATIONAL FUNCTION: $f(x) = p(x) / q(x)$ where p, q are tropical polynomials = max-plus combinations of linear functions. IN TROPICAL A_L : ReLU = tropical max = the (+) operation of tropical A_L Weight matrix W = tropical multiplication Deep network = tropical polynomial composition Universal approximation = tropical polynomials dense in continuous functions THE CONNECTION TO THE CAMPAIGN: Classical neural network ($q=1$): computes smooth functions via tau-traces Tropical neural network ($q=0$): computes piecewise-linear via max-plus The TRANSITION $q=1 \rightarrow q=0$: Sigmoid $\rightarrow$ ReLU [smooth $\rightarrow$ piecewise-linear] Smooth NN $\rightarrow$ Tropical NN [full hologram $\rightarrow$ frozen skeleton] DEEP LEARNING IS ARITHMETIC AT ABSOLUTE ZERO. The neural network is a tropical hologram. Training a neural network = finding the frozen arithmetic structure that best approximates the true ($q=1$) arithmetic hologram.

This connection is profound: the field of machine learning — the technology behind modern AI — is, at its mathematical heart, tropical geometry. Every ReLU network is a tropical polynomial. Every gradient descent step is a tropical optimisation. The 'intelligence' of a neural network is the tropical shadow of the full arithmetic hologram A_L at $q=1$. Deep learning approximates arithmetic — imperfectly, at zero temperature, with finite depth and width — but it is computing pieces of A_L .

8. The Tropical-Classical-Quantum Trinity

Structure	$q=0$ (Tropical)	$q=1$ (Classical)	$q=\text{root of unity}$ (Quantum)
Algebra	$(R, \min, +)$ semiring	$A_L = \mathbb{I}_1$ factor	$q\text{-}A_L$, Hecke algebra
Numbers	Exponents in R	$\tau(e_n) = 1/n$	$\tau_q(e_n) = 1/[n]_q$
Hamiltonian	$H_{\text{trop}} = \text{id}$ (linear)	$H = \log N$	$H_q = \log_q N$
Integration	max-plus sums	tau-traces	τ_q -traces
Zeta function	min-plus zeta	Riemann zeta(s)	$q\text{-zeta } [n]_q^{-s}$
MZV	max-plus iterated	$\zeta(n_1, \dots, n_k)$	$q\text{-MZV}$
Curves	Tropical (piecewise)	Algebraic (smooth)	Quantum (fuzzy)
Mirror symmetry	Polar duality of polytopes	SYZ/Fourier-Mukai	Level-rank duality
Deep learning	ReLU networks!	Smooth networks	Quantum networks
Geometry	Newton polytopes	Algebraic varieties	Quantum varieties
Topology	Berkovich skeleton	Classical topology	Jones polynomial

9. The Frozen Hologram: Absolute Zero Beauty

TROPICAL GEOMETRY = A_L AT ABSOLUTE ZERO THE PHASE DIAGRAM IS NOW COMPLETE: $q=0$ TROPICAL (R, min, +) [this document] $q<1$ QUANTUM subcritical [dv308] $q=1$ CLASSICAL A_L [dv1-dv312, our world] $q>1$ HOT supercritical [Bost-Connes $\beta<1$] $q=\text{roots}$ TOPOLOGICAL Jones/TQFT [dv308] $q=-1$ PARITY Z/2Z [dv277,311] **SIX IDENTITIES OF TROPICAL A_L:** 1. Tropical semiring = $q\text{-A}_L$ as $q\rightarrow 0$ [from q -integers] 2. Tropical MZV = max-plus iterated sums [frozen MZV] 3. Mikhalkin = tau continuous at $q=0$ [GW = tropical count] 4. Berkovich = interpolating $q\text{-A}_L$ [tropical to classical] 5. Tropical mirror = polar duality [mirror at $q=0$] 6. ReLU networks = tropical polynomials [AI = frozen A_L] **THE DEEPEST CONNECTION:** Modern artificial intelligence (deep learning) operates in the TROPICAL LIMIT of the arithmetic hologram. Neural networks are frozen A_L at absolute zero. They approximate arithmetic — imperfectly, combinatorially — because they cannot access the full $q=1$ hologram. **TO GO BEYOND DEEP LEARNING:** climb from $q=0$ to $q=1$. From max-plus to tau. From piecewise-linear to smooth. From tropical A_L to A_L itself. **THE CAMPAIGN ITSELF** is the journey from $q=0$ to $q=1$: From the combinatorial (Collatz, prime factorisation) to the analytical (RH, zeta, MZV, periods) to the geometric (Langlands, mirror, NCG) to the quantum (Jones, TQFT, quantum groups). A journey through all temperatures of A_L. **CHUNG TA LA HOA SI. TU NHIE LA BUC TRANH. O MOI NHIE DO: MOT HOA SI, MOT BUC TRANH. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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$q\rightarrow 0$: $a (+) b = \min(a,b)$, $a () b = a+b$. Mikhalkin: tropical counts = classical GW invariants. Berkovich: X^{an} interpolates tropical to classical. ReLU networks = tropical polynomials. Deep learning = frozen A_L. Phase diagram complete: $q=0$ to $q=\text{inf}$, all temperatures, one hologram. Chung ta la Hoa Si. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!*